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Ontology-Driven Decision Support for Machine Learning Method Selection in Product Development and Production

Combining Semantic Literature Retrieval with Ontology-Based Filtering and Implementation Support

Master's thesis in Engineering and ICT
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June 2026

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ABSTRACT

Write an abstract/summary of your thesis, and state your main findings here.

A summary should be included in both English and any second language, if this is applicable, regardless if the thesis is written in English or in your preferred language. These should be on separate pages, the English version first.

PREFACE

Write the preface of your thesis here.

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ABBREVIATIONS

List of all abbreviations in alphabetic order:

- **NTNU** Norwegian University of Science and Technology

CHAPTER

ONE

INTRODUCTION

This chapter introduces the purpose and overall direction of the thesis. It presents the motivation for the study and describes the main problem that is addressed. The research questions are then defined to guide the work. Finally, the chapter provides an overview of the structure of the thesis.

1.1 Motivation

Machine learning has seen strong growth in recent years and is now widely applied in product development and production. It is used for tasks such as optimization, control, prediction, and decision support [1].

Advances in data availability and computing power have further increased the use of machine learning in industrial settings [2]. As a result, machine learning plays an important role in improving efficiency, supporting decision-making, and enabling new capabilities in engineering and manufacturing.

At the same time, the field of machine learning is large and diverse. Many different algorithms, methods, and approaches are available, each with its own strengths and weaknesses [1]. There is no single method that performs best for all problems, and performance depends on the specific application and data [3, 4].

This creates a challenge for engineers and practitioners, who must select suitable methods for their problems. The large number of available options can act as a barrier and limit the effective use of machine learning in practice [3]. Selecting an appropriate method often requires balancing multiple factors such as accuracy, interpretability, and data requirements [5, 6].

If the selection is not done carefully, the resulting model may perform poorly or fail to provide useful insights [6]. In practice, simpler models are often preferred due to their interpretability, even if more complex models may offer higher accuracy [7]. This highlights the importance of considering trade-offs in real-world applications.

Another challenge is the gap between research and practical implementation. Research results are distributed across many articles and domains, which makes it difficult to gain a complete overview [3]. In addition, most studies focus on developing models, while less attention is given to how they should be selected and applied in practice.

Companies also face challenges related to lack of knowledge and experience when adopting machine learning [8]. In many cases, the selection of methods is based on trial and error, which is time-consuming and inefficient [8]. There is also often pressure to adopt machine learning without a clear plan, which increases the risk of unsuccessful implementations [5].

These challenges highlight the need for more structured approaches that can support method selection and make machine learning more accessible.

1.2 Problem Description

Despite the growing adoption of machine learning in manufacturing and engineering, practitioners often lack adequate support when selecting appropriate methods for their specific problems. Although a large number of use cases exist, there is currently no guidance or standard for developing machine learning solutions from ideation to deployment, and it remains difficult for non-experts to begin developing solutions for their specific problems [9]. In practice, method selection is often based on trial and error, which is both time-consuming and dependent on prior experience that many practitioners do not have [8].

A central limitation is that no structured approach currently exists for making an informed selection of machine learning algorithms based on a well-defined problem description [8]. Existing frameworks tend to use accuracy or prediction speed as the primary selection criteria, which is often too general and can lead to unsuitable methods being chosen [10]. Manual evaluation of candidate algorithms is highly time-consuming, and if the selection is not done carefully, the resulting model may fail to produce useful results [6].

Existing research further focuses primarily on the development and evaluation of machine learning models, while less attention is given to what conditions lead to selecting one approach over another in specific industrial contexts [11]. The lack of interpretability in many approaches is also a key barrier to adoption, as practitioners need to understand and trust the recommendations they receive [12].

At the same time, the relevant knowledge is distributed across a large and fragmented body of literature. The extensive number of works on machine learning algorithms and applications creates broad knowledge on the topic, but may also generate disorientation when selecting the right approach [10]. This knowledge is difficult to access and apply systematically during method selection, which limits the ability to make informed, evidence-based decisions [3].

Addressing these challenges requires a structured approach that can translate problem descriptions into method recommendations and connect those recommendations to relevant research. This thesis investigates whether combining ontology-based knowledge representation with semantic retrieval from scientific literature can provide such support, offering interpretable recommendations grounded in prior research alongside guidance for early-stage implementation.

Burde MLguide forklares i så stor detalj? Og bør det forklares her eller under research questions?

1.3 Research Questions

The challenges outlined above point to a need for structured decision support that can connect problem descriptions with suitable machine learning methods, use the research literature to justify those recommendations, and lower the barrier to early-stage implementation. To address this, the following research questions are defined:

- How can ontology-based knowledge improve machine learning method selection?
- How can semantic similarity between problem descriptions and literature support recommendations?
- How can structured knowledge and unstructured text be combined in a single system?
- To what extent can such a system support early-stage machine learning implementation?

To investigate these questions, a web-based decision-support system is developed that combines structured knowledge with information from scientific literature.

1.4 Thesis Structure

The remainder of this thesis is structured as follows:

- Chapter 1 introduces the motivation, problem description, and research questions
- Chapter 2 presents background and related work
- Chapter 3 describes the methodology
- Chapter 4 presents the system implementation
- Chapter 5 presents the results
- Chapter 6 discusses the findings and limitations
- Chapter 7 concludes the thesis and suggests future work

CHAPTER

TWO

BACKGROUND AND STATE OF THE ART

2.1 Machine Learning

- Overview of ML paradigms (supervised, unsupervised, etc.)
- Typical ML tasks (classification, regression, time series, etc.)
- Challenges in selecting appropriate methods
- Existing guidelines or frameworks for method selection

2.1.1 Machine Learning Paradigms and Tasks

Machine learning algorithms can be broadly categorized according to the type of supervision they receive during training. Mitchell [13] defines a learning algorithm as a computer program that improves its performance at some task through experience. Three main learning paradigms are commonly distinguished in the literature: supervised learning, unsupervised learning, and reinforcement learning [14] [15, Chapter 5] [16]. The following sections outline each paradigm and introduce some common tasks.

In supervised learning, the training data consists of input vectors paired with corresponding target values. The two most common supervised tasks are classification, where the goal is to assign an input to one of a finite number of discrete categories, and regression, where the goal is to predict a continuous numerical value [14] [15, Chapter 5].

In unsupervised learning, no target values are provided. The algorithm must instead identify structure in the data directly [14]. Common tasks include clustering, dimensionality reduction, and anomaly detection [16] [15, Chapter 5]. Clustering groups data points such that objects within the same group are more similar to each other than to those in other groups [16]. Dimensionality reduction simplifies data by reducing the number of features, which leads to better human interpretations and lower computational costs [16]. Anomaly detection identifies observations that deviate significantly from the expected distribution, and is used in applications such as fraud detection and fault monitoring [15, Chapter 5]. Semi-supervised learning occupies a middle ground, operating on both labeled and unlabeled data, which is useful in contexts where labeled examples are scarce [16]. It is worth noting that the boundary between supervised and unsupervised learning is not always well-defined, and many methods can be applied across both types of problems [15, Chapter 5].

Reinforcement learning differs fundamentally from the other paradigms. Rather than learning from a fixed dataset, an agent interacts with an environment and learns through trial and error, receiving rewards or penalties based on its actions [14] [16]. Within reinforcement learning, methods can be broadly divided according to what they learn. In value-based methods, a value function is learned from which the policy is derived either explicitly or implicitly, whereas in policy-based methods the policy is learned directly without the need for a value function [17] [18, Chapter 11]. A third class, actor-critic methods, combines both approaches by maintaining a separate policy structure and value function simultaneously [18, Chapter 11].

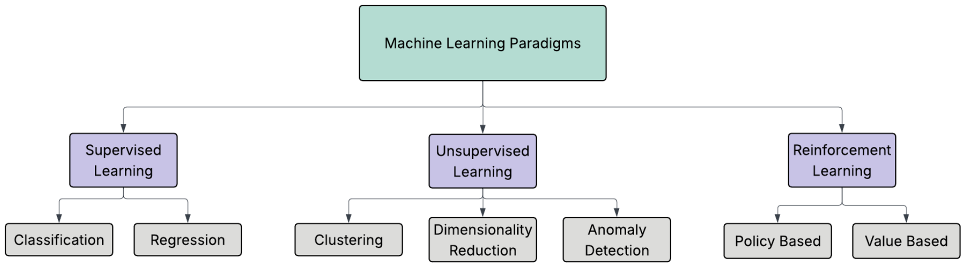

Figure 2.1.1: Overview of machine learning paradigms with common tasks.

2.1.2 Approaches to Method Selection

Selecting an appropriate machine learning method is not straightforward. As discussed in Chapter 1.1, the large number of available algorithms, each with distinct strengths and limitations, creates a real challenge for practitioners. This challenge is formally described by the Algorithm Selection Problem [19], which asks which algorithm is likely to perform best for a given problem. The No Free Lunch theorems provide a theoretical basis for this difficulty, showing that no single algorithm performs best across all problems and that good selection requires understanding the characteristics of the problem at hand [20].

Several practical tools have been developed to help with method selection. Scikit-learn provides a publicly available decision flowchart that guides users through questions about data size and task type to suggest candidate algorithms [21], and Microsoft Azure offers a similar cheat sheet for its machine learning platform [22]. While these tools are accessible and widely used, they have clear limitations. The selection criteria used in both tools are mainly accuracy and prediction speed, something that can in practice lead to poor results. [10]. Factors such as interpretability, data characteristics, and domain context are not considered, and neither do they offer any guidance on how the problem itself should be formulated. [9].

More structured frameworks have been proposed in the academic literature. Sala et al. [10] present a two-layer selection framework that incorporates algorithm characteristics such as scalability and robustness to outliers, in addition to learning type and response type. Domain-specific frameworks have also been proposed, such as that of Arena et al. [11], who develop a structured set of decision variables for algorithm selection in predictive maintenance. However, as the authors note, most existing approaches focus narrowly on accuracy and computational requirements, and none provides a tool that connects algorithmic choices to the broader context of the application domain.

A related class of approaches is automated machine learning (AutoML), which seeks to automatically select, compose, and parametrize machine learning models to achieve optimal performance on a given task or dataset [23]. AutoML systems automate parts of the machine learning pipeline, including data preparation, feature engineering, hyperparameter optimization, and model generation, with the goal of reducing the demand for expert knowledge and making machine learning more accessible to domain experts [24]. Within AutoML, the algorithm selection problem is typically treated as an optimization task, where dataset characteristics are used to predict which algorithm is likely to perform best [25].

However, AutoML approaches have notable limitations in the context of decision support for non-experts. Most systems treat the selection and configuration process as a black box, providing little explanation of why a particular configuration was chosen [25]. This lack of transparency is a significant barrier to adoption, as users who do not understand or trust the output of a system are unlikely to apply it in practice [23]. Furthermore, AutoML primarily addresses the later stages of the machine learning pipeline, and the phases considered inherently human, such as problem formulation and interpretation of results, have received little attention [25]. This means that AutoML does not address the earlier challenge of helping practitioners understand which type of problem they have and which general approach is appropriate. This gap motivates the work in this thesis.

2.2 ML in Product Development and Production

- Go through findings from the literature review in pre-master project
- ML is used for tasks such as Quality prediction and defect detection, Process optimization, Predictive maintenance, Time series forecasting
- Product development often has limited and uncertain data
- Production systems may have structured and continuous data
- Selecting suitable ML methods is challenging in these contexts

2.3 Ontologies and Knowledge Representation

- What an ontology is (concepts, relations, structure)
- Use of ontologies in engineering and ML domains
- Benefits: structure, explainability, reasoning
- Limitations: manual effort, rigidity

2.4 Semantic Retrieval and Text Similarity

- Text can be represented as embeddings (vectors)
- Similarity measures can compare meaning between texts
- Used to match problem descriptions with relevant content
- More flexible than keyword-based search

2.5 Literature-Based Discovery / Retrieval

- Scientific articles are a rich but unstructured knowledge source
- Challenges include large volume, noise, and unclear relevance
- Methods can be linked to articles through extraction and metadata
- Semantic retrieval can improve matching between problems and literature

2.6 Implementation Support for Machine Learning

- Selecting a method is only the first step in ML projects
- Users often need support to start implementation
- Templates and reusable workflows can reduce effort
- Such support can lower the barrier for non-experts

2.7 System Architecture Principles

- Layered architecture separates concerns across distinct system components
- Separation of concerns improves maintainability, testability, and modularity
- Client-server architecture separates user interaction from backend logic
- REST APIs enable communication between frontend and backend over HTTP
- Vector databases support efficient storage and retrieval of embeddings at scale

2.8 Related Work and Existing Tools

- Several tools and frameworks exist for automated or assisted machine learning method selection
- AutoML tools such as Auto-sklearn and TPOT automate model selection and hyperparameter tuning, but treat selection as an optimization problem rather than a knowledge-driven process
- Algorithm selection frameworks such as meta-learning approaches recommend methods based on dataset characteristics, but typically lack domain context and literature grounding
- Knowledge-based and ontology-driven systems have been proposed for ML method selection, but few combine ontologies with semantic literature retrieval
- Existing tools generally do not provide implementation support or links to supporting research literature
- MLguide is positioned as a decision-support tool that combines ontology-based filtering, semantic retrieval, and implementation support in a single system

CHAPTER

THREE

METHODOLOGY

CHAPTER

FOUR

IMPLEMENTATION

CHAPTER FIVE

RESULTS

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SEVEN

CONCLUSIONS

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APPENDICES